

ORIGINAL ARTICLE

Effect of ciprofloxacin and *in vitro* gut conditions on biofilm of *Escherichia coli* isolated from clinical and environmental sources

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Abstract

Aim: This study aimed at characterizing the biofilm-forming ability of drug-resistant and sensitive *Escherichia coli* under *in vitro* gut conditions and in the presence of ciprofloxacin.

Methods and Results: 153 *E. coli* isolates comprising 80 from clinical and 73 from environment source were studied for their ability to form biofilm under control and *in vitro* simulated gut conditions. The integrity of preformed biofilm on exposure to ciprofloxacin was assessed. Expression of biofilm-associated genes was analysed using qPCR. A high degree of resistance was observed in clinical isolates with a concomitant prevalence of *bla*_{TEM}. Bile, pH and low temperature enabled the *E. coli* biofilm to resist the effect of ciprofloxacin. Clinical isolates of *E. coli* formed strong biofilms in *in vitro* gut conditions following exposure to high concentration of ciprofloxacin. The expression of biofilm genes varied between different gut conditions viz., presence of bile, pH and low temperature, included in this study.

Conclusions: This study demonstrates the importance of *papC* and *csgA* for maintaining the biofilm integrity upon antibiotic exposure. *Escherichia coli* form biofilm as a survival strategy to adapt to the conditions in their environment irrespective of their drug resistance status.

Significance and Impact of the Study: The study provides an understanding of the effect of different parameters of the gut conditions during infection and the effect of antibiotic on survival and biofilm-forming ability of clinical and environmental *E. coli* isolates. It further suggests that bacteria resort to biofilm formation as one of the mechanisms to adjust to alterations in gut conditions and once the biofilm is formed, it requires high concentration of ciprofloxacin to eradicate it.

KEYWORDS

antimicrobial drug resistance, bacterial adhesion, ciprofloxacin, *E. coli*, *in vitro* gut condition

INTRODUCTION

Escherichia coli is a common inhabitant of the intestine of humans and animals. Not only is it a common gut microbiota, but also has a beneficial role in the maintenance of intestinal flora (Yan & Brent Polk, 2004). When

a person is immunocompromised or the intestinal barriers are violated, nonpathogenic micro-organisms can become opportunistic pathogens and cause infection by acquiring specific virulence factors by horizontal gene transfer of mobile genetic elements. In addition, pathogenicity islands, and plasmids confer resistance, enable

them to adopt new-niche and the ability to cause disease (Kaper et al., 2004). *Escherichia coli* is the causative agent of urinary tract infections (UTI), parenteral infections, neonatal meningitis, prostatitis, and Gram-negative bacteraemia (Sekhar et al., 2017). Enteric *E. coli* causes gastroenteritis but rarely any disease outside the gut. However, extra-intestinal pathogenic *E. coli* (ExPEC) strains can disseminate from the gut and colonize the circulatory system, central nervous system and urinary tract (Wiles et al., 2008). Among the several extra-intestinal infections caused by *E. coli*, UTIs are the most common in humans with 40% women and 12% men being affected at least once in their lifetime (González et al., 2017). The risk factors related to UTI are mainly associated with the ability of *E. coli* to form biofilms. This leads to persistence of *E. coli* in the vaginal reservoir or bladder epithelium (Sanchez et al., 2013). It is well recognized that biofilms are bacterial communities embedded in the exopolysaccharide matrix that protect the bacteria from antibiotics and make it 1000 times more resistant than its planktonic stage (Donlan and Costerton, 2002; Kotian et al., 2020). They are produced by environmental stress such as the presence of bile, variation in osmolarity, iron uptake, pH, and temperature during infection by altering the bacterial gene expression related to virulence factors, surface molecules, and metabolic status of the organism (Rossi, Cimdins, et al., 2018). Ciprofloxacin is the widely used second-generation fluoroquinolone antibiotic to treat the UTI caused by *E. coli* and other Gram-negative pathogens (Dong et al., 2019). Due to its wide usage, it is usually found in hospital effluents, wastewater treatment plants, and also in rivers, rendering bacterial pathogens resistant to ciprofloxacin (Hassan et al., 2020). Wastewater from the hospitals disseminate these antibiotic-resistant bacteria to humans and animal populations through various modes such as food and water wherein the pathogens like *E. coli* persist as biofilms (Maheshwari et al., 2016). Many factors are known to influence biofilm formation in bacterial pathogens. However, the effect of gut environmental conditions and antibiotic such as ciprofloxacin on *E. coli* biofilm is still unclear. Hence the present study aims at identifying the gut environmental factors that have tremendous effect on *E. coli* biofilm when exposed to the most commonly used fluoroquinolone, ciprofloxacin.

MATERIALS AND METHODS

Bacterial strains

Escherichia coli isolated from clinical ($n = 80$) and environmental ($n = 73$) sources were used in the study. Cultures from the institutional repository were revived using tryptone soy broth (TSB).

Antimicrobial resistance pattern and minimum inhibitory concentration

All the isolates were subjected to antimicrobial susceptibility testing by Kirby Bauer's disc diffusion method for 10 antimicrobial agents viz ampicillin (10 µg), imipenem (10 µg), cefotaxime (30 µg), ciprofloxacin (5 µg), levofloxacin (5 µg), nalidixic acid (30 µg), nitrofurantoin (300 µg), tetracycline (30 µg), chloramphenicol (30 µg) and cotrimoxazole (25 µg) (HiMedia Laboratories Pvt Ltd). The minimum inhibitory concentration (MIC) to ciprofloxacin was performed using MIC E-strips (HiMedia Laboratories Pvt Ltd). The experiments were performed following the Clinical and Laboratory Standards Institute (CLSI, 2016) guidelines. The isolates were classified as susceptible, intermediate and resistant. The ATCC *E. coli* (25922) was used as the quality control strain.

Characterization of antimicrobial resistance and virulence genes

The isolates showing phenotypic resistance to antibiotics were screened for the presence of corresponding resistance determinants and virulence genes using PCR. Briefly, PCR was carried out in a 30 µl reaction containing 22.1 µl of sterile water, 3 µl of 10× assay buffer (100 mM Tris-HCL (pH 9), 1.5 mM MgCl₂, 50 mM KCL and 1% gelatin), 0.6 µl of 200 mM of four dNTP mix (dATP, dGTP, dCTP and dTTP), 10 pmol of each forward and reverse primers (Tables 1 and 2), 1.0 U of *Taq* DNA polymerase and 2 µl of template DNA in a thermal cycler (Bio-Rad) (Table S1).

Biofilm formation by *E. coli*

All the 153 *E. coli* isolates were studied for their ability to form biofilm using crystal violet assay (Stepanović et al., 2000). Briefly, 200 µl of TSB and 20 µl of overnight grown culture was added to the wells of the microtitre plate and incubated at 37°C for 24 h. Following this, the wells were decanted and washed thrice with sterile water. 220 µl of 1% crystal violet was added to each well and incubated at ambient temperature ($27^{\circ} \pm 2^{\circ}\text{C}$) for 15 min. The contents of the wells were emptied and excess of crystal violet removed by washing the wells with sterile water. The crystal violet bound to the biofilm was solubilized with 33% glacial acetic acid. The OD was measured at 630 nm using UV spectrophotometer (Thermo Scientific) and true OD values were obtained by subtracting the control value. Classification of isolates was done as follows: $\text{O.D} \leq \text{O.D.c}$ = non biofilm producer, $\text{O.D. c} < \text{O.D} \leq (2 * \text{O.D.c})$ = weak biofilm producer, $(2 * \text{O.D.c}) < \text{O.D} \leq (4 * \text{O.D.c})$ = moderate biofilm

TABLE 1 Oligonucleotides used for the detection of antibiotic resistance genes in *Escherichia coli*

Antibiotic	Primer	Primer sequence	Size(bp)	Reference
Ampicillin	<i>bla</i> _{TEM}	F-TCACCCAGAAACGCTGGTG R-TCCGCCTCCTCCAGTCTA	569	Colomer-Lluch et al. (2011)
Cefotaxime	CTXM1-F3	F-GACGATGTCACTGGCTGAGC	499	Pitout et al. (2004)
CTX-M group I	CTXM1-R2	R-AGCCGCCGACGCTAATACA		
CTX-M group II	TOHO1-2F	F-GCGACCTGGTTAACTACAATCC	351	
	TOHO1-1R	R-CGGTAGTATTGCCCTTAAGCC		
CTX-M group III	CTXM825F	F-CGCTTTGCCATGTGCAGCACC	307	
	CTXM825R	R-GCTCAGTACGATCGAGCC		
CTX-M group IV	CTX-M 914F	F-GCTGGAGAAAAGCAGCGGAG	474	
	CTXM914R	R-GTAAGCTGACGCAACGTCTG		
Imipenem	<i>bla</i> _{NDM-1}	F-GGTTTGGCGATCTGGTTTTTC R-GGATGGCTCATCACGATC	621	Nordmann et al. (2011)
Tetracyclines	<i>tetA</i>	F-GCGCCTTTCTTTGGGTTCT R-CCACCCGTTCCACGTTGTTTA	494	Ma et al. (2007)
	<i>tetB</i>	F-CAGTGCTGTTGTTGTCTTAA R-GCTTGAATACTGAGTGTTAA	571	
	<i>tetD</i>	F-GCAAACCATTACGGCATTCT R-GATAAGCTGCGCGGTAAAAA	546	
	<i>tetE</i>	F-TATTAACGGGCTGGCATTTC R-AGCTGTCAGGTGGGTCAAAC	544	
	<i>tetG</i>	F-GCTCGGTGGTATCTCTGCTC R-CAAAGCCCCTTGCTTGTTAC	550	
Chloramphenicol	<i>cat 1</i>	F-AACCAGACCGTTCAGCTGGAT R-CCTGCCACTCATCGCAGTAC	549	
	<i>cat 2</i>	F-AACGGCATGATGAACCTGAA R-ATCCCAATGGCATCGTAAAG	547	
	<i>cat 3</i>	F-ATCGGCATCGGTTACCATGT R-ATCCCCTTCTTGCTGATATT	531	
Sulphonamides	<i>sul I</i>	F-TTTCCTGACCCTGCGCTCTAT R-GTGCGGACGTAGTCAGCGCCA	425	
	<i>sul II</i>	F-CCTGTTTCGTCCGACACACAGA R-GAAGCGCAGCAGCCGCAATTCAT	435	
	<i>sul III</i>	F-ATGAGCAAGATTTTGAATCGT R-CTAACCTAGGGCTTTGGATATTT	792	
Plasmid-mediated quinolone resistance genes	<i>qnrA</i>	F-ATTTCTCACGCCAGGATTTG R-GATCGGCAAAGGTTAGGTCA	516	Robicsek et al. (2006)
	<i>qnrB</i>	F-GATCGTGAAAGCCAGAAAGG R-ACGATGCCTGGTAGTTGTCC	469	
	<i>qnrS</i>	F-ACGACATTCGTCAACTGCAA R-TAAATTGGCACCCCTGTAGGC	417	
	<i>qepA</i>	F-CGTGTTGCTGGAGTTCTTC R-CTGCAGGTACTGCGTCATG	403	

producer, $4 \times \text{O.D.c} < \text{O.D} = \text{strong biofilm producer}$ (Stepanović et al., 2000). All the isolates were also checked for biofilm production under independent *in vitro* gut conditions which included bile (3%), salt (0.3 M NaCl), high iron (100 μM FeCl_3), low iron (0.2 mmol l^{-1} of 2,2'-dipyridyl), temperature (25°C), pH (5.8) (Kröger et al., 2013) and high temperature (39°C) (Zorc et al., 2005).

Effect of ciprofloxacin on the integrity of preformed biofilms

The concentration 1000 $\mu\text{g ml}^{-1}$ of the antibiotic ciprofloxacin used in the study was based on its bioavailability in human urine (Lorian, 2005). All the isolates with strong and moderate biofilm-forming ability under control and

TABLE 2 Oligonucleotides used for the detection of virulence genes in *Escherichia coli*

Virulence genes	Function	Primer sequence	Size(bp)	Reference
<i>FimH</i>	Adhesion	F-GGTGAAAACGACGCGGTATT R-GATGTCACCGTTACTCTGCC	152	This study
<i>PapC</i>	Pilli associated with pyelonephritis	F-GCCAGTGCCGTTGAGTTTAA R-TCTGTAACGATGCGGGAGAA	157	
<i>IroN</i>	Catecholate siderophore	F-TCCGACGATGATAATGACGA R-ATTGACGCCAGGCATTTTAC	185	
<i>iutA</i>	Ferric aerobactin receptor	F-ACGTCTTTCCGTGGCATATC R-TTAAGCCCGTAATCGTCGTC	207	
<i>PAICFT073</i>	Pathogenicity-associated island marker	F-CTGGGGATCCAGTCCTACAA R-CGGCCAGATAAAAGGAATGA	198	
<i>BcsA</i>	Catalytic subunit of the cellulose synthase	F-CCGTCGGTGGATATCTTTGT R-GCAAACCTGGCGAAACTCTTTC	152	
<i>CsgA</i>	Major subunit of curli fimbriae	F-GCCACCACCGAACTGTTTAA R-TGCCGTAACCTGACTTGA	178	
<i>MotA</i>	Motility protein A	GCTGGAATAGAGCGTTTGC R-GTGGGGACTTTCCTCGGTAT	186	
<i>agn43</i>	Autoaggregation factor	F-ACACCACACTCACCCAGAAA R-AGAGTGACATTCTGTGGGGTC	160	

the various *in vitro* conditions were selected. The biofilm was grown in triplicate as described previously with slight modification. After washing with sterile water to remove nonadherent cells, the well was filled with 180 μ l of media and 1000 μ g ml⁻¹ of antibiotic solution. The plates were incubated for 24 h at 37°C (except the plates wherein different temperatures of incubation for the *in vitro* simulated gut conditions were used), contents of the wells emptied, washed with sterile water followed by crystal violet staining (Nair et al., 2020). The absorbance was measured at 630 nm to see the effect of antibiotic on preformed biofilm.

RNA extraction by TRIzol

The expression studies of biofilm genes of selected isolates under control and *in vitro* conditions were carried out. Biofilm of the individual isolates was developed on a coverslip in six-well cell-culture plate in TSB for 24 h at 37°C under control and *in vitro* gut conditions following which, the nonbiofilm-forming cells were removed by gentle rinse and wash. The coverslip was removed and the biofilm formed on it was washed with sterile TSB. By addition of TRIzol (Invitrogen), the bacterial cells were lysed and incubated for 5 min at room temperature (27 $\pm$ 2°C) to allow dissociation of nucleoprotein complex. To the mixture, 200 μ l of chloroform (Merck Lifescience) was added and vortexed for 15 s followed by centrifugation at 12,000 g at 4°C for 15 min. The upper aqueous phase was transferred to a fresh tube and RNA precipitated from it by using 0.5 ml isopropyl alcohol followed by centrifugation at 12,000 g for 10 min at 4°C.

The pellet was washed with 75% ethanol and resuspended in 30 μ l of nuclease-free water. The concentration and purity were determined in a spectrophotometer (Bio Spectrometer) and the RNA stored at -80°C for later use.

CDNA synthesis

Reverse transcription was carried out according to the manufacturer's instructions (iScript cDNA synthesis, Biorad). 20 μ l reaction mixture containing 4 μ l of 5 $\times$ iScriptTM reaction mix, 1 μ l reverse transcriptase enzyme, 1 μ g of RNA template and nuclease-free water was subjected to reverse transcription. The reaction protocol of priming at 25°C for 5 min, reverse transcription at 46°C for 20 min and reverse transcriptase inactivation for 1 min at 95°C were performed in a thermocycler (Bio-Rad).

Expression analysis of biofilm-associated genes

Real-time PCR was performed to quantitate the expression of biofilm genes viz., *bcsA* (cellulose synthesis), *csgA* (curli production), *papC* (pili associated with pyelonephritis), *fimH* (type 1 fimbriae), *motA* (motility protein A) and *agn43* (antigen 43). 16S rRNA was used as housekeeping gene and for normalization of data for qPCR which was carried out using Sso AdvancedTM universal SYBR green super mix (BioRad). The appropriate concentration of primers was determined and melt curve analysis was performed

to check the amplification of untargeted regions. Relative quantification of all the genes was calculated using the $2^{-\Delta\Delta Ct}$ formula. 9 μ l of the real-time PCR master mix containing 5 μ l of SYBR green super mix and 200 nmol l⁻¹ of forward and reverse primers was distributed to each well of the 96-well plate and 1 μ l of cDNA was used as a template. The reaction was performed in real-time PCR detection system (CFX96TouchTM, Bio-Rad) with the cycling conditions as follows: initial denaturation at 95°C for 3 min followed by 40 cycles of denaturation at 95°C for 10 s, annealing at 52°C for 20 s and final extension at 72°C for 20 s. Data acquisition was performed by Bio-Rad CFX manager at the end of each elongation step. All the obtained expression values are presented as graphs (heat maps) by using GraphPad Prism software version 8.0 for Windows (GraphPad Software).

RESULTS

Antimicrobial resistance pattern

Among the 80 clinical isolates, 60 (75%) were MDR, 20 (25%) were resistant to at least one antibiotic. None of the isolates was susceptible to the antibiotics used for testing in this study. Of the 80 clinical isolates tested for sensitivity to β -lactam antibiotics, 78 (97.50%) were resistant to ampicillin, 76 (95%) to cefotaxime and 42 (52.50%) to imipenem. Among quinolones/fluoroquinolones included for sensitivity testing, 73 (91.25%), 59 (73.75%) and 51 (63.75%) isolates were resistant to nalidixic acid, ciprofloxacin and levofloxacin respectively. The MIC to ciprofloxacin was $>256 \mu\text{g ml}^{-1}$ in 49 (61.25%) isolates. Among protein synthesis inhibiting antibiotics, 43 (53.75%) were resistant to tetracycline. With the 73 environmental *E. coli* isolates, 28 (38.4%) were MDR and 21 (28.75%) were resistant to at least one antibiotic while 24 (32.85%) were susceptible to all the antibiotics included in the study. Among the three β -lactam antibiotics used for testing against 73 environmental isolates, 35 (47.95%), 32 (43.84%) and 8 (10.95%) were resistant to ampicillin, cefotaxime and imipenem respectively. Among the quinolones/fluoroquinolones used for testing, 41 (56%), 27 (37%) and 5 (7%) isolates showed resistance to nalidixic acid, ciprofloxacin and levofloxacin respectively. The MIC to ciprofloxacin was $>32 \mu\text{g ml}^{-1}$ in 22 (30.13%) isolates. 16 (22%) environmental isolates were resistant to nitrofurantoin, the protein synthesis inhibitor (Figure S1).

Characterization of antimicrobial resistance genes and virulence genes

Isolates that showed phenotypic resistance to an antibiotic were selected for the detection of antimicrobial

resistance genes of the corresponding class. 57 (73%) of 78 clinical and 16 (45.71%) of 35 environmental isolates showing resistance to ampicillin carried *bla*_{TEM}. Four sets of primers were used to determine cefotaxime resistance. Of the 76 clinical and 32 environmental isolates resistant to cefotaxime, 59 (77.63%) clinical and 18 (56.25%) environmental were positive for *bla*_{CTX-M} from the CTX-M-I group demonstrating the increasing presence of CTX-M-1-like β -lactamases. Likewise, of 42 clinical and 8 environmental isolates showing resistance to imipenem, 6 (14.2%) clinical and 2 (25%) environmental isolates harboured *bla*_{NDM-1}. Of 43 clinical isolates resistant to tetracycline, *tetA* was present in 17 (39%) and 8 (18.60%) carried *tetE*. None of the isolates carried *tetB*, *tetD* and *tetG*. Of the 12 environmental isolates resistant to tetracycline, 6 were positive for *tetA*, 2 harboured *tetB* and only one carried *tetD*. All the isolates were negative for *tetE* and *tetG*. One isolate carried three tetracycline resistance genes viz, *tetA*, *tetB* and *tetD*. Fluoroquinolone resistance was seen in 79 (98.75%) clinical and 41 (51.25%) environmental isolates. Of the four genes, *qnrA*, *qnrB*, *qnrS* and *qepA*, associated with plasmid-mediated quinolone resistance (PMQR), *qnrS* was found to be highly prevalent and harboured by 8 (10%) clinical and 9 (21.95%) environmental isolates. *qnrB* was seen in 6 (7.5%) and 4 (9.75%) isolates from clinical and environmental sources respectively. *qnrA* was detected only in 2 (2.5%) clinical isolates while none of the environmental isolates was positive. Except one environmental isolate, none of the isolates carried *qepA*. Among the *sul* genes encoding sulphonamide resistance, the majority of the isolates carried *sulI* with 37 (72.54%) clinical and 16 (69.5%) from environmental source being positive. 17 (33.3%) clinical and 11 (47%) environmental isolates were positive for *sulII*. However, none of the tested clinical isolates harboured *sulIII* while 3 (13%) environmental isolates harboured *sulIII*. With respect to 8 clinical isolates showing resistance to chloramphenicol, 5 (62.5%) carried *catA1* and 3 (37.5%) were positive for *catA2*. However, of nine chloramphenicol-resistant environmental isolates, only one harboured both *catA1* and *catA2*. It was interesting that none of the isolates from both clinical and environmental sources harboured *catA3*.

A study of the 5 virulence genes in the study showed *fimH* to be prevalent in 62 (77.5%) clinical and 58 (79.45%) environmental isolates. 41 (51.25%) clinical and 7 (9.5%) environmental isolates were positive for *iutA*. *papC* was found in 32 (40%) clinical isolates but was absent in all the environmental isolates. The pathogenicity-associated island gene *PAICT073* was present in 52 (65%) clinical and 3 (4.1%) environmental isolates. The less prevalent iron siderophore gene *iroN* was present in 13 (16.25%) clinical and 4 (5.47%) environmental isolates (Table 3).

TABLE 3 Resistance determinants and virulence genes detected by PCR

Resistance genes	Corresponding antibiotic	Clinical isolates phenotypic (+)	Clinical isolates genotypic (+)	Environmental isolates phenotypic (+)	Environmental isolates genotypic (+)
<i>bla</i> _{TEM}	Ampicillin	78	57(73%)	35	16(45.71%)
CTX-M group-I	Cefotaxime	76	59(77.63%)	32	18(56.25%)
CTX-M group-II			1(1.31%)		1(3.25%)
CTX-M group-III			1(1.31%)		1(3.25%)
CTX-M group-IV			2(2.62%)		8(25%)
<i>bla</i> _{NDM-1}	Imipenem	42	6(14%)	8	2(25%)
<i>tetA</i>	Tetracycline	43	17(40%)	12	6(50%)
<i>tetB</i>			0		2(16.6%)
<i>tetD</i>			0		1(8.3%)
<i>tetE</i>			8(18.60%)		0
<i>tetG</i>			0		0
<i>qnrA</i>	Plasmid-mediated quinolone resistance	79	2(2.5%)	41	0
<i>qnrB</i>			6(7.5%)		4(9.75%)
<i>qnrS</i>			8(10%)		9(21.95%)
<i>qepA</i>			0		1(2.43%)
<i>sul</i> I	Sulphonamides	51	37(72.54%)	23	16(69.5%)
<i>sul</i> II			17(33.3%)		11(47.8%)
<i>sul</i> III			0		3(13.04%)
<i>cat</i> 1	Chloramphenicols	8	5(62.5%)	9	1(11%)
<i>cat</i> 2			3(37.5%)		1(11%)
<i>cat</i> 3			0		0
Virulence gene		Clinical isolates(<i>n</i> = 80)		Environmental isolates (<i>n</i> = 73)	
<i>fimH</i>		62(77.5%)		58(79.45%)	
<i>PapC</i>		32(40%)		0	
<i>PAICT073</i>		52(65%)		3(4.109%)	
<i>iroN</i>		13(16.25%)		4(5.47%)	
<i>iutA</i>		41(51.25%)		7(9.5%)	

Biofilm formation by *E. coli*

Of 80 clinical *E. coli* studied for biofilm-forming ability under control conditions, 8 (10%) were strong, 9 (11.25%) moderate, 28 (35%) weak, and 35 (43.75%) isolates were nonbiofilm former. With 73 environmental isolates, 4 (5%) were strong, 3 (4%) moderate, 21 (29%) weak and 45 (62%) were nonbiofilm formers. When the isolates were subjected to *in vitro* gut conditions such as bile, the number of strong biofilm formers increased to 24 (30%) clinical and 9 (13%) environmental isolates. In the presence of 0.3 M NaCl, 5 (6.25%) clinical and 5 (7%) environmental isolates formed strong biofilm. Under high iron condition, 6 (7.5%) clinical and 4 (5%) environmental isolates were strong biofilm formers and in low iron condition, 7 (8.75%) clinical and 4 (5%) environmental isolates formed strong biofilm. At pH

5.8, 23 (28.75%) clinical and 5 (7%) environmental isolates formed strong biofilm. At a lower temperature of 25°C, 34 (42.5%) clinical and 15 (21%) environmental isolates formed strong biofilms whereas at a high of 39°C, only 11 (14%) clinical and 4 (5.47%) environmental isolates formed strong biofilm (Figure 1a,b) (Tables S2 and S3).

Effect of ciprofloxacin on the integrity of preformed biofilm

On exposure to ciprofloxacin under control condition or under some *in vitro* conditions such as NaCl, high and low iron, high temperature, none of the clinical isolates that were earlier recorded as strong biofilm retain their original character. However, when isolates were grown

under certain *in vitro* conditions of gut like bile, pH and low-temperature conditions the following results were obtained. 4 (16.66%) isolates in presence of bile, 4 (17.39%)

at pH of 5.8, 18 (54.54%) at low temperature of 25°C that formed strong biofilm retained their biofilm strength even after ciprofloxacin exposure (Figure 2a,b). None of the

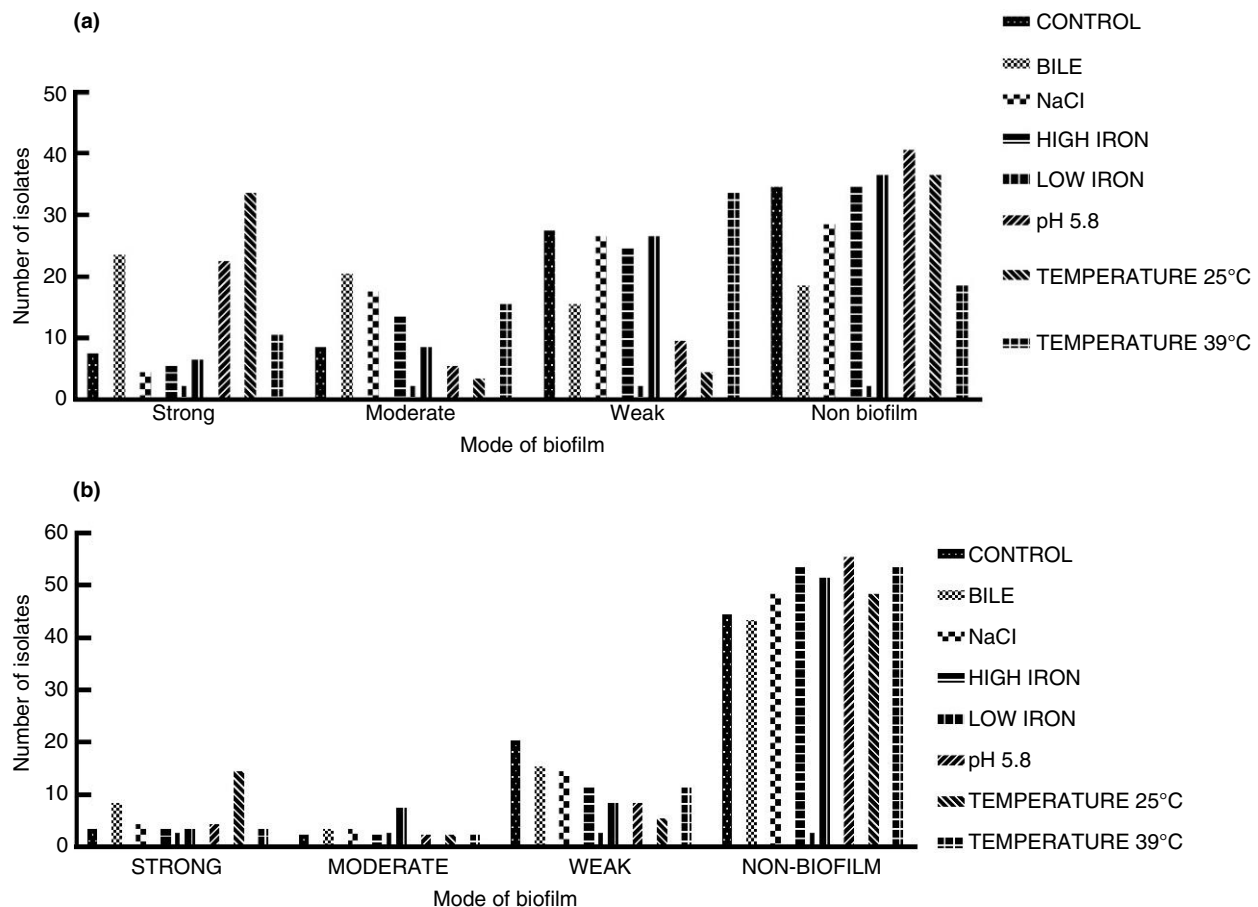


FIGURE 1 Effect of *in-vitro* gut conditions. (a) Biofilm formation by clinical isolates under *in-vitro* gut conditions. (b) Biofilm formation by environmental isolates under *in-vitro* gut conditions

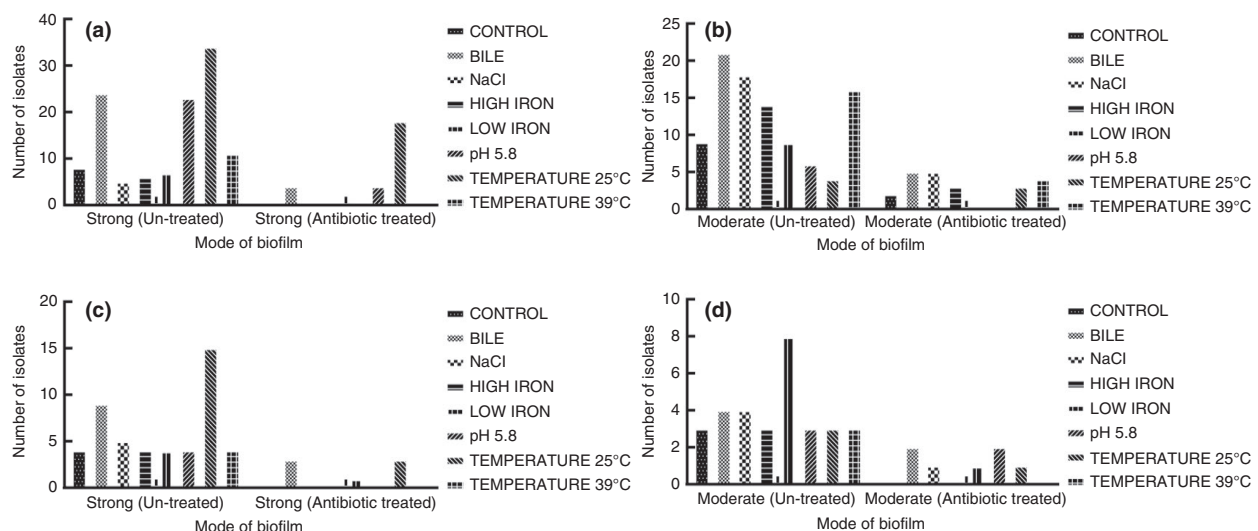


FIGURE 2 Effect of ciprofloxacin on pre-formed biofilm of *Escherichia coli*. (a) Effect of ciprofloxacin on clinical strong biofilm. (b) Effect of ciprofloxacin on clinical moderate biofilm. (c) Effect of ciprofloxacin on environmental strong biofilm. (d) Effect of ciprofloxacin on environmental moderate biofilm

strong biofilm-forming environmental isolates under control, NaCl, high iron and pH, high-temperature conditions remained strong after ciprofloxacin exposure. However, strong biofilms of 3 (33%) environmental isolates formed under conditions of bile, 1 (25%) in presence of low iron, and 3 (23.07%) at a low temperature of 25°C remained strong biofilm formers even after ciprofloxacin exposure (Figure 2c,d).

Biofilm gene expression of *E. coli*

Relative expression of six different genes involved in biofilm production was studied on exposure to different *in vitro* conditions. PCR for the six genes revealed their presence in clinical isolates; however, two genes, *papC* and *iutA*, were absent in environmental isolates. Fold change in all the transcripts levels was logarithmically expressed. Among clinical isolates, the cellulose producing gene *bcsA* was upregulated in J111 (LT), J226 (HI, LI, pH, bile), J176 (LT), J43 (HT), J123 (bile & pH) and J250 (NaCl). The gene responsible for curli production, *csgA* was upregulated in J111 (LT), in J226 (HI, pH, LT, bile), J176 (LT), J43 (HT), J123 (pH), J250 (NaCl) and MU4 (LI). The *papC* which correlates with increased biofilm development was upregulated in all the conditions except in J111 (NaCl, pH), J123 (pH) and MU4 (HI & LI). In the environmental isolates *bcsA* was upregulated in 197 (HI & LI, LT&HT), 136

(NaCl) and 104k (HT) while *csgA* was upregulated in isolates 197 (bile, HI & LI, HT), 136 (HI & NaCl) and 104k (HT) (Figures 3 and 4) (Tables 4 and 5).

DISCUSSION

Antibiotic resistance is a major problem globally presenting challenge to treatment of bacterial infections. In the present study, 75% of the clinical isolates were MDR while 25% were resistant to at least one antimicrobial agents. None of the clinical isolates were sensitive to all the antibiotics tested. Prevalence of resistance was high towards β -lactams, sulphonamides and fluoroquinolones. One of the many sources that contribute to antibiotic resistance is hospital wastewater which contain antibiotics and/or ARDs which can generate resistant strains through horizontal transfer of ARDs (Garcia et al., 2007). Likewise, resistance was also found in environmental isolates to β -lactams, sulphonamides and fluoroquinolones. 38% of the isolates were MDR and 28.75% were resistant to one or more antibiotics indicating environment as a hotspot of resistant bacteria. Among the quinolones, high degree of resistance to nalidixic acid (56%), followed by ciprofloxacin (37%) was observed and found to be in agreement with Abbasi and Ranjbar (2018).

*bla*_{TEM} was the common ampicillin resistance gene found in 73% of clinical and 50% of environmental

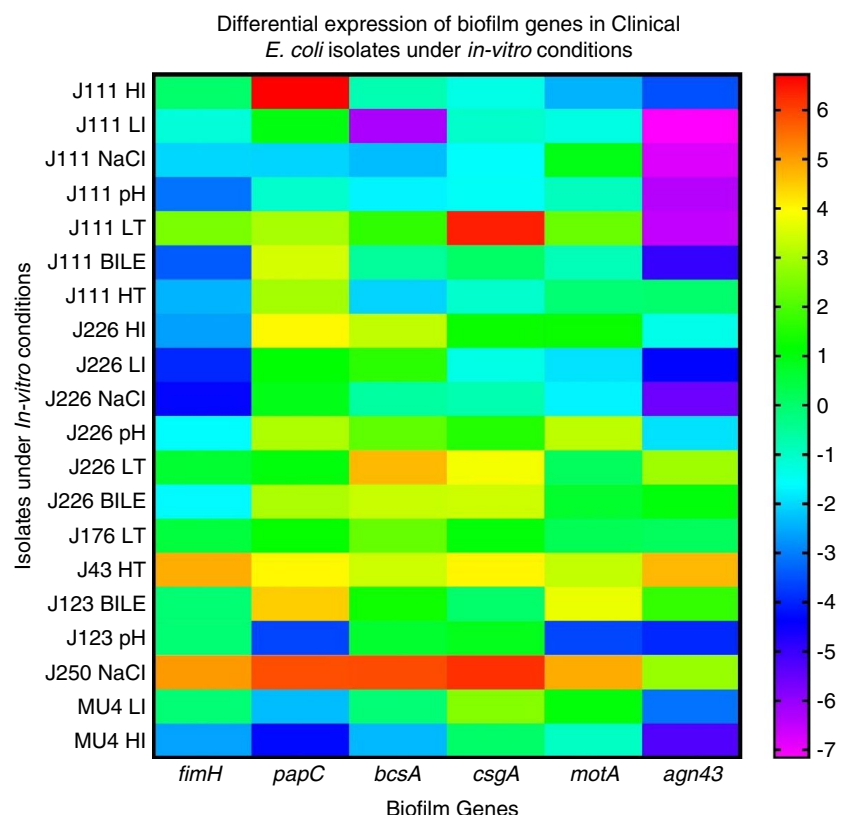


FIGURE 3 The expression level of biofilm genes in clinical isolates are presented using fold change values transformed to log₂ format compared to control. The log₂ (fold-change values) and the color scale are shown right to the heat map

FIGURE 4 The expression level of biofilm genes in environmental isolates are presented using fold change values transformed to log₂ format compared to control. The log₂ (fold-change values) and the color scale are shown right to the heat map

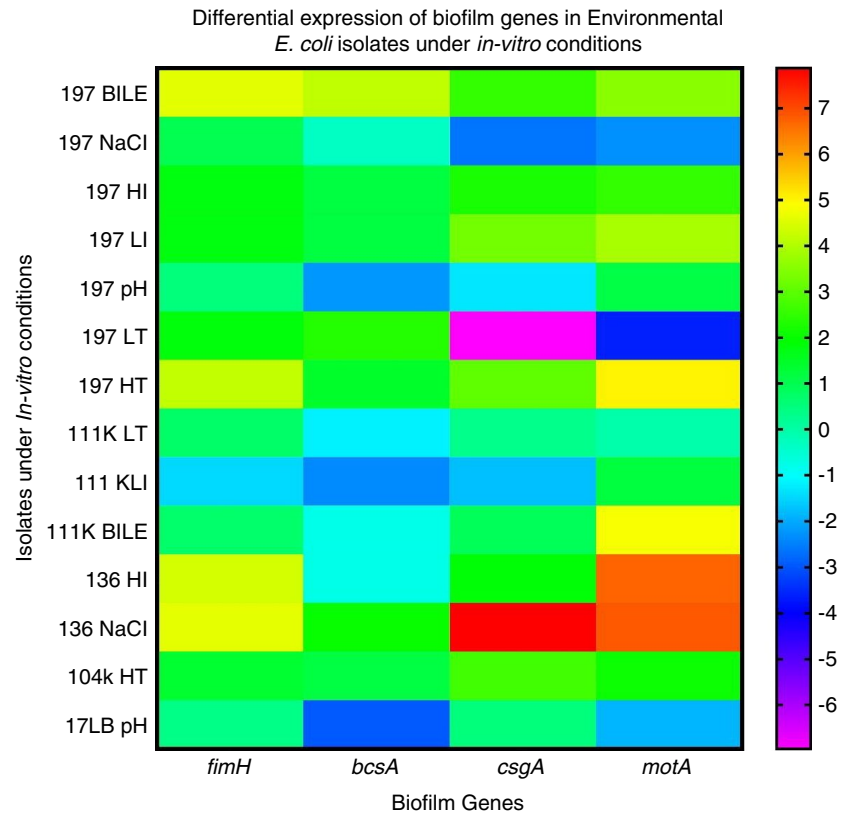


TABLE 4 Relative expressions of biofilm genes of clinical isolates under in vitro gut conditions

Isolates	<i>FimH</i>	<i>PapC</i>	<i>bcsA</i>	<i>csgA</i>	<i>motA</i>	<i>agn43</i>
J111 HI	0.01 ± 0.3	6.71 ± 0.5	-0.76 ± 0.2	-1.33 ± 0.2	-2.44 ± 0.1	-3.51 ± 0.05
J111 LI	-1.19 ± 0.06	0.97 ± 1.2	-6.25 ± 0.00	-1.04 ± 0.07	-1.29 ± 0.12	-7.1 ± 0.00
J111 NaCl	-2.04 ± 0.05	-2.07 ± 0.10	-2.32 ± 0.05	-1.58 ± 0.01	0.93 ± 0.49	-6.79 ± 0.01
J111 pH	-3.13 ± 0.03	-1.05 ± 0.4	-1.70 ± 0.03	-1.51 ± 0.04	-0.91 ± 0.15	-6.38 ± 0.00
J111 LT	2.47 ± 0.2	2.97 ± 0.5	1.65 ± 0.4	6.37 ± 0.8	2.31 ± 1.2	-6.50 ± 0.00
J111 BILE	-3.39 ± 0.03	3.47 ± 1.2	-0.51 ± 0.3	0.07 ± 0.2	-0.85 ± 0.24	-4.96 ± 0.02
J111 HT	-2.41 ± 0.01	2.94 ± 0.2	-2.07 ± 0.01	-1.06 ± 0.13	-0.08 ± 0.17	-0.02 ± 0.77
J226 HI	-2.65 ± 0.1	4.00 ± 0.6	3.26 ± 0.2	1.26 ± 0.2	1.24 ± 0.2	-1.38 ± 0.2
J226 LI	-3.39 ± 0.04	1.14 ± 0.8	1.60 ± 0.2	-1.32 ± 0.1	-1.91 ± 0.01	-4.35 ± 0.0
J226 NaCl	-4.32 ± 0.03	0.91 ± 0.9	-0.57 ± 0.07	-0.72 ± 0.1	-1.72 ± 0.01	-5.57 ± 0.2
J226 pH	-1.63 ± 0.1	3.05 ± 0.8	2.18 ± 0.7	1.53 ± 0.7	3.22 ± 0.3	-1.92 ± 0.0
J226 LT	0.64 ± 1.0	1.03 ± 0.6	4.68 ± 0.2	3.83 ± 0.2	0.15 ± 0.1	2.88 ± 0.2
J226 BILE	-1.64 ± 0.2	3.04 ± 0.2	3.34 ± 0.2	3.39 ± 0.5	0.66 ± 0.02	1.05 ± 0.5
J176 LT	0.49 ± 0.3	1.19 ± 0.4	2.26 ± 0.9	1.07 ± 0.2	0.24 ± 0.02	0.14 ± 0.2
J43 HT	4.83 ± 0.8	4.01 ± 0.3	3.40 ± 0.1	4.02 ± 0.5	3.29 ± 0.3	4.69 ± 0.3
J123 BILE	-0.08 ± 0.02	4.47 ± 0.2	1.31 ± 0.5	0.02 ± 0.5	3.72 ± 0.5	1.71 ± 0.05
J123 pH	-0.07 ± 0.1	-3.64 ± 0.5	0.66 ± 0.4	0.86 ± 0.1	-3.62 ± 0.3	-3.96 ± 0.2
J250 NaCl	5.04 ± 0.5	5.85 ± 0.5	5.90 ± 0.2	6.19 ± 0.1	4.87 ± 0.5	2.78 ± 0.6
MU4 LI	-0.09 ± 0.2	-2.33 ± 0.1	-0.11 ± 0.3	2.58 ± 0.2	1.06 ± 0.2	-3.13 ± 0.0
MU4 HI	-2.60 ± 0.3	-4.29 ± 0.2	-2.38 ± 0.01	0.04 ± 0.6	-0.94 ± 0.3	-5.26 ± 0.0

isolates. Of the four different CTX-M β -lactamase groups, a large number of isolates from both the sources were positive for CTX-M-1- β -lactamase (First group). This is in

agreement with earlier studies wherein the dominance of group 1 CTX-M was observed in extra-intestinal *E. coli* isolates (Kim et al., 2016; Rohit et al., 2019). A low degree of

TABLE 5 Relative expressions of biofilm genes of environmental isolates under in vitro gut conditions

Isolates	<i>FimH</i>	<i>bcsA</i>	<i>csgA</i>	<i>motA</i>
197 BILE	4.57 ± 1.1	4.15 ± 2.4	2.51 ± 1.2	3.52 ± 1.2
197 NaCl	1.0 ± 0.6	−0.32 ± 0.45	−2.64 ± 0.09	−2.32 ± 0.1
197 HI	1.75 ± 0.6	1.17 ± 0.3	2.19 ± 0.32	2.50 ± 1.2
197 LI	1.75 ± 1.14	1.16 ± 0.89	3.27 ± 1.1	3.90 ± 1.3
197 pH	0.49 ± 0.1	−2.25 ± 0.21	−1.32 ± 0.26	1.10 ± 0.3
197 LT	1.79 ± 1.3	2.34 ± 1.2	−6.96 ± 0.00	−3.64 ± 0.03
197HT	4.18 ± 1.4	1.43 ± 0.21	3.03 ± 0.5	5.02 ± 1.2
111K LT	0.74 ± 0.3	−1.18 ± 0.12	0.31 ± 0.8	−0.04 ± 0.03
111K LI	−1.47 ± 0.03	−2.39 ± 0.2	−1.75 ± 0.18	1.2 ± 0.9
111K BILE	0.71 ± 0.3	−0.79 ± 0.3	0.9 ± 0.6	4.8 ± 0.2
136 HI	4.38 ± 0.2	−0.76 ± 0.2	1.85 ± 0.2	6.7 ± 0.1
136 NaCl	4.59 ± 0.3	1.99 ± 0.1	7.87 ± 0.3	6.83 ± 0.5
104K HT	1.36 ± 0.8	1.15 ± 0.9	2.69 ± 0.2	2.05 ± 0.2
17LB pH	0.35 ± 0.4	−2.97 ± 0.06	0.5 ± 0.2	−1.87 ± 0.1

resistance observed towards protein synthesis inhibitors when compared to β -lactams and fluoroquinolones might be due to the limited use of such antibiotics as reported in developed countries (Erjavec et al., 2007).

sulI coding for sulphonamide resistance was more prevalent than *tetA* coding for tetracycline resistance. This conforms to the report by Deekshit et al. (2012) wherein majority of the tetracycline-resistant nontyphoidal *Salmonella* strains isolated from seafood harboured *tetA* in a transferable plasmid and *sulI* in class 1 integron (Deekshit et al., 2012). In this study, all the phenotypic-resistant isolates did not always carry genotype of the corresponding antibiotic. This could be due to the presence of other mechanisms of resistance or presence of resistance determinants that were not included in the study. In *E. coli*, resistance to sulphonamides can arise either due to chromosomal mutation or by the acquisition of alternative dihydropteroate synthase (DHPS) genes like *sulI*, *sulII* and *sulIII* (Adnan et al., 2017). 72% of clinical isolates carried *sulI*, 33% carried *sulII* while *sulIII* was absent in all the isolates. Among environmental isolates, 69% carried *sulI*, 47% carried *sulII* and 13% harboured *sulIII*, and this study was in accordance with the results of Arabi et al. (2015). Another class of antibiotics that is widely used to control bacterial infections are fluoroquinolones. Among PMQR genes studied, *qnrA* was absent in environmental strains while *qepA* was absent in clinical strains. The prevalence of PMQR genes was less than 10% in isolates from both the sources. Pereira et al. (2007) found only one isolate harbouring *qnr* among 144 ciprofloxacin-resistant isolates (Pereira et al., 2007). This could be due to other resistance mechanisms such as QRDR mutations or over-expression of efflux pumps in fluoroquinolone-resistant

strains (Jazeela et al., 2019; Rezazadeh et al., 2016; Shetty et al., 2019).

Antibiotic resistance genes not only help the bacteria to survive exposure to antibiotics but also help to adapt to conditions in the body and to overcome innate mechanisms of killing or arresting the growth of pathogens. To establish an infection, virulence factors are required for *E. coli*. In this study, five virulence genes of uropathogenic *E. coli* were included. The gene *fimH* that encodes type 1 fimbriae, an important virulence factor, promotes adhesion, invasion and biofilm growth (Zamani & Salehzadeh, 2018). 78% of clinical and 79% of environmental isolates harboured *fimH*. Another adhesion gene, the pili associated with pyelonephritis (*papC*) was detected in 40% of clinical isolates but was absent in environmental isolates. This is in contrast to other studies wherein high proportion of *papC* was encountered in avian pathogenic and other environmental isolates of *E. coli* (Enany et al., 2019; Titilawo et al., 2015).

After successful attachment to the infection site, iron plays an important role in the adaptation of bacteria to the surrounding environment. Iron deficiency weakens the adaptive abilities by modifying the capsules (Simões et al., 2007). To adapt to such iron limiting conditions, bacteria need to develop mechanisms that enable them to sequester iron from the sites of infection. Production of siderophores that have high affinity for iron and helps uptake is one such mechanism (Sarowska et al., 2019). Siderophore-related genes, *iroN* was recorded in 16% of clinical and 5% environmental isolates and *iutA* was detected in 51% clinical and 10% environmental isolates. Naves et al. (2008) also reported *iroN* and *iutA* in the majority of the clinical isolates that they studied (Naves et al., 2008). Another

virulence gene, *CFT073*, a genetic marker for uropathogenic *E. coli*, was detected in 65% clinical and 4% environmental isolates. The study provides evidence that isolates from hospital and environmental sources are hotspots for antibiotic resistance and virulence genes that help adaptation to the gut environment once they find entry through contaminated water or food (Biswal et al., 2014).

Additionally, as a survival strategy to overcome the unfavourable environment of the gut including the pressure of antibiotics, the bacteria form biofilms (Kotian et al., 2020). In this study, the biofilm-forming ability was higher in clinical isolates than in environmental isolates. Most of the clinical isolates were strong biofilm formers under *in vitro* gut conditions such as the presence of bile. This corroborates the report by Kotian et al. (2020) who observed that majority of the seafood associated nontyphoidal *Salmonella* showed strong biofilm formation in the presence of bile (Kotian et al., 2020). *Escherichia coli*, being a gut commensal with some being pathogenic types, have evolved to survive in the presence of bile salts using bile as a signal to modulate the production of adhesive structures that further help in biofilm formation (Rossi, et al., 2018). At pH 5.8, most of the weak biofilm-forming clinical *E. coli* isolates transformed to strong biofilm formers. Both clinical and environmental isolates of *Vibrio vulnificus* could form strong biofilms at neutral pH (Çam & Brinkmeyer, 2019). However, environmental isolates showed decreased biofilm-forming ability at pH 5.8. In this study, the *in vitro* gut condition like 0.3 M NaCl did not have a discernible effect on biofilm formation by either clinical or environmental isolates. Some studies have shown that increase in osmolarity results in decreased biofilm formation (Prigent-Combaret et al., 2001). However, *Burkholderia pseudomallei*, a nosocomial pathogen formed strong biofilm when grown in a media supplemented with NaCl (Pumirat et al., 2017). On the role of temperature, the number of strong biofilm formers increased in both clinical and environmental isolates at 25°C. This corroborates the report by Mathlouthi et al. (2018) wherein they showed that low temperature favoured strong biofilm formation by *E. coli* (Mathlouthi et al., 2018).

Further when the isolates were subjected to *in vitro* gut conditions, nonbiofilm formers were able to form biofilm. The transition from nonbiofilm to biofilm formers, or weak biofilm former to moderate or strong biofilm former was higher in clinical isolates. The expression of biofilm genes plays a crucial role in biofilm formation by enteric pathogens (Gerstel & Römling, 2003; Pratt & Kolter, 1998; Wood et al., 2006). Isolates that changed from moderate to strong biofilm formers and nonbiofilm to weak under different *in vitro* gut conditions were selected for biofilm gene expression studies. *papC* was consistently upregulated followed by *bcsA*, *csgA* and *motA* under *in vitro* gut

conditions for the clinical isolates showing moderate and strong biofilm. Since most of the clinical *E. coli* isolates were UPEC, *papC* is a common pili associated gene critical for adhesion, biofilm formation and pathogenesis of ascending UTI (Gunardi et al., 2021; Naves et al., 2008; Soto et al., 2007). However, in environmental isolates, since *papC* was absent, the type I pili-associated *fimH* was consistently upregulated followed by *motA*, *csgA* and *bcsA*. *fimH* is required for surface attachment and recent studies by Zuberi et al. (2017) demonstrated that suppression of *fimH* using CRISPRi decreased biofilm formation (Zuberi et al., 2017). The architecture of biofilm is influenced by motility which plays a critical role in biofilm formation (Pratt & Kolter, 1998; Wood et al., 2006). In a study conducted by Pratt and Kolter (1998), paralysed flagella (*motA*) resulted in defective biofilm formation. Wood et al. (2006) also highlighted the importance of *motA* in mature biofilm formation (Pratt & Kolter, 1998; Wood et al., 2006). These results suggest that *E. coli* uses both common and specific genes for the development of biofilm. However, in this study, a greater number of clinical isolates formed biofilms than environmental isolates. This could be due to the presence of additional genes such as *papC*, *agn43* which were absent in environmental isolates.

The effects of antibiotic on preformed biofilm varied between different *in vitro* gut conditions. Sub-MIC of ciprofloxacin has previously been shown to reduce biofilm formation (Dong et al., 2019) while another study showed contrary to the above in that it induced biofilm formation (Rafaque et al., 2020). However, during infection, pathogens are exposed to varied conditions in the gut; therefore, the effect of antibiotics under different *in vitro* gut conditions need better understanding. In this study, the majority of the clinical isolates showed an MIC of greater than 256 µg ml⁻¹ while environmental isolates showed an MIC of greater than 32 µg ml⁻¹ for ciprofloxacin. In sub-MIC concentration of ciprofloxacin, the bacteria were not completely eradicated and therefore have an opportunity to modulate their virulence traits and develop resistance. Since biofilms help the bacteria to withstand high level of antibiotic, a high concentration of ciprofloxacin was used to study the effect of antibiotic on preformed biofilm under different *in vitro* gut conditions on selected strong and moderate biofilm formers. Majority of the selected clinical isolates continued to be strong biofilm formers at low temperature (18 out of 34), followed by pH (4 out of 23) and bile (4 out of 24). However, in environmental isolates only few remained strong after antibiotic exposure. This might be due to the differential expression of biofilm genes in both clinical and environmental isolates under *in vitro* gut conditions. For instance, a strong biofilm-forming clinical isolate J111 and a moderate biofilm-forming isolate J226 did not undergo alteration in their biofilm characteristic

even after ciprofloxacin exposure in low temperature. This could possibly be explained by the high expression of all the biofilm genes under low temperature in these isolates (Figure 3; Table 4). Likewise, these isolates remained strong and moderate in bile after exposure to ciprofloxacin, which could be attributed to the up-regulation of all the biofilm genes except *fimH* in J226 and up-regulation of only *papC* and *csgA* in J111. The role of *papC* and *csgA* in maintaining biofilm architecture and strength after antibiotic exposure is obvious in this study.

In conclusion, the study reflects the increased biofilm-forming ability of clinical isolates in comparison to the environmental on subjecting to *in vitro* physiological gut conditions like bile, pH and temperature. The role of virulence genes such as *papC*, *iroN* in clinical isolates could be a major contributing factor (Naves et al., 2008). There was a positive correlation between *iutA* and biofilm formers and is in contrast to the observation by Naves et al. (2008) who found no significance difference between the presence and absence of *iutA* with strong and weak biofilm formers (Naves et al., 2008). Most clinical isolates showed high resistance to ciprofloxacin and had *gyrA* and *parC* mutations while very few isolates were positive for plasmid-mediated quinolone resistance determinants (Jazeela et al., 2019; Shetty et al., 2019). Most of the clinical isolates were able to transform from nonbiofilm former to biofilm former under *in vitro* gut conditions. Further, clinical isolates were able to form strong biofilms due to the presence of additional genes such as *papC* and *agn43*. To adapt to *in vitro* gut condition, bacteria form biofilm, and high concentration of antibiotic is required to eradicate them. Though ciprofloxacin was able to eradicate biofilm formed by most of the isolates, low temperature followed by bile and pH could still resist the effect of high ciprofloxacin concentration. The results are a pointer to the fact that infection-related *in vitro* gut conditions included in this study possibly help a pathogen such as *E. coli* to become more virulent coupled with resistance to antibiotics by forming strong biofilms *in vivo*. An antibiotic susceptible human pathogen *in vitro* can therefore become resistant *in vivo* by forming strong biofilm. Thus, the mere determination antimicrobial susceptibility of a bacterial pathogen will not be adequate to select appropriate antimicrobial therapy for the treatment of severe infections. Hence it is also essential to understand the biofilm formation and virulence characteristics of the bacterial pathogens for the successful antimicrobial therapy.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Vankadari Aditya performed the experiments, wrote the manuscript; Akshatha Kotian contributed in data revision and performed the experiment; Sreya Saikrishnan, Dr. Anusha Rohit, Dr. Divyashree M and Dr. Indrani Karunasagar contributed in data revision; Dr. Deekshit VK conceptualized the ideas and experimental design of the study.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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